

Lab 12 – Recursion

Aim

This lab is intended to give you practice with using recursion.

Resources

Your Week 12 lecture notes will be useful: follow the Recursion Recipe!

Tips:

1. Recursion can be challenging, and will be easier if you first pause and think about it!
2. This lab will use no for/while loops. No loops at all, only recursion!

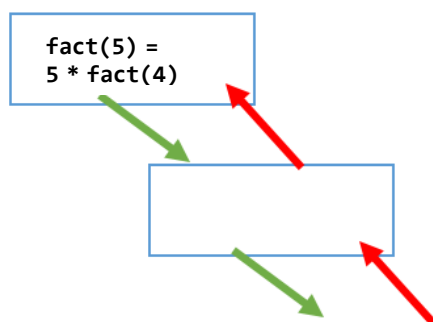
Recursion – Factorial

Your Week 12 lecture notes introduce many problems, including Factorial and Fibonacci.

- Consider the Factorial problem. A suitable set of test cases would look like:

Input Integer	Expected Answer	Your Answer	Pass Test?
0	1		
1	1		
2	2		
3	6		
4	24		
5	120		

- We can solve this using loops, but this week, we are going to use recursion.
- Review the week 12 lecture notes, create a method **fact** to solve this problem.
- What is the process through the program? It can help to draw the process. Starting with an input of 5, how many times does it call the method? Complete the diagram:



Student Money 1

Given a class of students, we wish to find out how much money there is in the class. We will not use close-form multiplication or any loops here, only recursion. Let us start simple: we assume that each student has 7 RMB. This leads to the following test cases:

Number Students	Expected Answer	Your Answer	Pass Test?
0	0		
1	7		
2	14		
3	21		
4	28		
5	35		
6	42		
7	49		

- Create a recursive method **studentMoney1** that calls itself.
- First, what are the base cases? What are the simplest possible cases?
- How can you then solve it recursively? Draw a diagram of the process.

Student Money 2 – Now with richer students

Given a class of students, we wish to find out how much money there is in the class. We will not use any loops here, only recursion. Here, most students have 7 RMB, but every 3rd student has 11 RMB.

- Student-1 has 7 RMB, student-2 has 7 RMB, student-3 has 11 RMB, student-4 has 7 RMB, student-5 has 7 RMB, student-6 has 11 RMB, and so on.
- First, what are your test cases? Fill in the table:

Number Students	Expected Answer	Your Answer	Pass Test?
0			
1			
2			
3			
4			
5			
6			
7			

- Create a method **studentMoney2** to solve this problem (Hint: use the mod integer operator). Test all your test cases to make sure it solves the problem.

Christmas Coming – Recursion with Strings

Christmas is coming, and we want to create an overexcited program. The method **christmasComing** will receive a string, and will recursively check each character as follows. If an "s" is found, then it will replace the "s" with "Santa".

- For example, the method call:
`System.out.println(christmasComing("christmasiscoming"));`
will produce the output: "chriSantatmaSantaiSantacoming"
- Create some test cases. Think about some example problems that might test if your program works:

Test case	Expected Answer	Your Answer	Pass Test?

- Create a recursive method to solve this! Think first about what the base case is, and what is the recursive step is.